Perihilar cholangiocarcinoma - practice decision chart

Draft v0.6 | MDT decision first | Evidence and technical feasibility | 4 October 2026

Drainage + staging + preserve biopsy and transplant routes

Resectable with adequate future liver remnant?

YES → Expert hepatobiliary resection pathway. Protect future options before biopsy or tumor-plane violation; manage drainage and remnant as planned.

NO ↓

Eligible for protocol-based transplant?

YES → Approved neoadjuvant therapy protocol, staging and transplant assessment. SBRT may be a protocol component; it is not independently curative evidence.

NO ↓

Outside resection / transplant pathway

Systemic therapy; selected definitive chemoradiation or palliative RT. Thermal ablation, Y90, TACE and HAI have no routine role for the hilar primary.

LOCAL MODALITY SELECTION AND CLINICAL EVIDENCE

Thermal ablation

No routine curative role at the hilar primary; major bile ducts and infiltrative growth limit safe margins.

Evidence gap

No established hilar-primary thermal-ablation trial supporting cure. Intraductal RFA is a different intervention.

No supporting trial PMID identified.

SBRT / ablative radiotherapy

Protocol component before transplant or selected definitive chemoradiation; SBRT alone has weak cure evidence.

Pilot transplant-protocol series

12 RT-treated (11 SBRT); 6 transplanted. Supports a combined pathway, not SBRT-alone cure.

PMID: 24115315

Y90 radioembolization

No routine role for the primary hilar tumor; exceptional liver-dominant recurrence only.

Evidence gap

No established hilar-primary Y90 trial supporting this role. MISPHEC studied intrahepatic CCA.

No supporting trial PMID identified.

TACE

No routine primary-tumor role; not a substitute for resection or transplant protocol.

Evidence gap

No established hilar-primary TACE trial supporting this role; do not extrapolate from iCCA.

No supporting trial PMID identified.

Hepatic arterial infusion (HAI)

No established routine curative role; do not extrapolate intrahepatic CCA pump data.

Evidence gap

No established hilar-primary HAI trial supporting cure. iCCA pump data are not direct evidence.

No supporting trial PMID identified.

Histotripsy

No tumor-specific curative evidence for a central hilar primary.

HOPE4LIVER | indirect evidence

Mixed HCC/metastasis cohort; no direct evidence for a central hilar primary.

PMID: 40201962

CONTRAINDICATIONS AND FEASIBILITY - SHARED ACROSS TUMOR TYPES

Thermal ablation

DO NOT PROCEED

Unsafe access or inability to cover tumor + margin without injuring central bile ducts, bowel or other critical structures; uncontrolled infection or uncorrectable bleeding risk.

CAUTION / PLANNING

Vessel-related heat sink (especially RFA); bilioenteric anastomosis/instrumentation increases abscess risk. Consider protection, another route or modality.

CIRSE standard | PMID: 32382856

SBRT / ablative radiotherapy

DO NOT PROCEED

Cannot meet normal-liver or GI dose limits; target cannot be reliably localized or respiratory motion controlled.

CAUTION / PLANNING

Poor hepatic reserve/decompensation, too little uninvolved liver, prior overlapping RT. Unmet central-biliary limit: lower dose or another modality. Adjacent bowel may need more fractions.

ASTRO guideline (HCC/iCCA) | PMID: 34688956

Y90 radioembolization

DO NOT PROCEED

Uncorrectable GI nontarget flow; lung exposure exceeds device/protocol limits despite mitigation; unsafe arterial access; clinical liver failure.

CAUTION / PLANNING

Limited reserve, prior liver RT, renal/contrast risk, biliary infection risk. Portal-vein thrombosis alone is not an automatic exclusion.

EANM guideline | PMID: 35146577

TACE

DO NOT PROCEED

Decompensated liver failure; severely impaired portal inflow without adequate collaterals; uncorrectable nontarget flow/access; uncontrolled infection.

CAUTION / PLANNING

Biliary-enteric communication/instrumentation raises abscess risk; renal/contrast risk and extensive burden. Portal-vein thrombosis needs individualized review.

CIRSE standard | PMID: 34694454

Hepatic arterial infusion (HAI)

DO NOT PROCEED

Pump HAI: uncorrectable extrahepatic perfusion or inadequate hepatic coverage; arterial occlusion, portal-vein obstruction/significant portal hypertension; poor reserve.

CAUTION / PLANNING

Must tolerate implantation and reliable refills/monitoring. Variant arteries may be correctable. Catheter HAIC and drug-specific rules differ.

Technical guidance | PMID: 36346892

Histotripsy

DO NOT PROCEED

Inadequate ultrasound visibility/acoustic window, obstructing ribs/lung/bowel gas, target outside device working range, or unsafe treatment-volume coverage.

CAUTION / PLANNING

Anesthesia/positioning and bleeding-risk review; critical-structure overlap requires expert review. Trial exclusions are not universal device contraindications.

Pivotal trial | PMID: 39225612; FDA DEN220087

DO NOT PROCEED = unresolved safety/feasibility barrier after attempted mitigation. CAUTION = relative risk requiring expert planning, not an automatic exclusion. This row is largely tumor-agnostic; disease extent, transplant strategy and regimen-specific eligibility remain in the roadmap. Device labeling and individual organ-at-risk constraints govern final planning.

REASSESS → preserve liver reserve and the next potentially curative option

Assess viable tumor, other disease, toxicity and reserve. Revisit complete-treatment options. If transplant is chosen, apply the full OPTN pathway; local response alone does not establish listing eligibility.

LC, local control; CR, complete response; OS, overall survival; PFS, progression-free survival; LPFS, local progression-free survival; RFS, recurrence-free survival. PMID links identify study evidence; selection thresholds also reflect the cited guidelines. No treatment is guaranteed curative.

NCCN Biliary Tract Cancers v1.2026, original PDF p22 (INTRA-D); perihilar pathway pp23-27. Full selection and allocation criteria: practice guideline and alignment worksheet.

LRT evidence update 23 October 2026 — institutional decisions pending

LT: approved neoadjuvant therapy and operative staging; OPTN does not mandate RT. Mass single and <3 cm for allocation. Avoid primary transperitoneal biopsy. Welling 24115315; Soliman 39936132; Hoogwater 37032423: observational protocol evidence, not equivalence.

Non-LT RT: selected MDT option. NCCN v1.2026 PDF p23 allows chemoradiation without prior systemic control; BSG supports SBRT/PBT after systemic therapy. No proven routine survival gain. STRONG: grade 3 cholangitis 5/6 (34439146); ABC-07: no PFS gain (ASCO2024 abstract4006).

Postoperative RT: NCCN PDF p24 includes R1 or node-positive disease; nonpreferred option for R0/node-negative disease. Systemic therapy and trial are preferred.

Endobiliary RFA: recent RCTs did not show survival/patency benefit (40619373,37652677). PDT: not recommended outside trials (30094069). I-125 stent plus HAIC: investigational, both arms had HAIC (39311927).

No established routine primary hilar role for Y90, TACE, thermal ablation or histotripsy. Drainage, infection control and bowel/bile-duct constraints govern feasibility.